

CO2-AT1-Scenario Based Programs

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QUESTIONS

Q1

Student Registration System
10 marks

Q2

E-Commerce Cart System
10 marks

Q3

Library Management System
10 marks

Q4

Employee Data Processing
10 marks

Q5

Online Voting System
10 marks

5/5 answered

Q1 Student Registration System

10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

- (a) Enumerate three Collection interfaces suitable for this system.
- (b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.
- (c) Write a Java program using HashSet and HashMap with Generics to:
- Add students
 - Map students to courses
 - Display data using Iterator

Your Code

```
1 import java.util.*;  
2 public class StudentRegistrationSystem {  
3     public static class Student {  
4         private int id;  
5         private String name;  
6         Student(int id, String name) {  
7             this.id = id;  
8             this.name = name;  
9         }  
10        public int getId() {  
11            return id;  
12        }  
13        public String getName() {  
14            return name;  
15        }  
16        public boolean equals(Object obj) {  
17            if (this == obj) return true;  
18            if (obj == null || getClass() != obj.getClass()) return false;  
19            Student student = (Student) obj;  
20            return id == student.id;  
21        }  
22        public int hashCode() {
```

Input

stdin input

Output

```
Student(id=101, name='Hasini')  
Enrolled Courses: [Math, CS,  
English]  
Student(id=102, name='balu')  
Enrolled Courses: [Physics,  
Chemistry]
```

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Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

- (a) List three Collection classes applicable for cart management.
- (b) Represent how cart items are stored using a List-based structure.
- (c) Write a Java program using ArrayList and Iterator to:
- Add items
 - Remove an item
 - Display all items

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 public class ECommerceCart {
4     public static void main(String[] args) {
5         ArrayList<String> cart = new ArrayList<>();
6         cart.add("Laptop");
7         cart.add("Mouse");
8         cart.add("Keyboard");
9         cart.add("Headphones");
10        System.out.println("Cart items after adding:");
11        displayItems(cart);
12        cart.remove("Mouse");
13        System.out.println("\nCart items after removing 'Mouse':");
14        displayItems(cart);
15    }
16    public static void displayItems(ArrayList<String> cart) {
17        Iterator<String> it = cart.iterator();
18        while (it.hasNext()) {
19            System.out.println(it.next());
20        }
21    }
22 }
```

Run

Save

Input

stdin input

Output

```
Cart items after adding:
Laptop
Mouse
Keyboard
Headphones
```


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10 marks

5/5 answered

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- (a) Name three Map implementations in Java.
- (b) Design a schema-like representation using Map for storing ISBN and book details.
- (c) Write a Java program using HashMap to:
- Insert book details
 - Retrieve a book
 - Display all books

Your Code

```
1 import java.util.*;
2 class Book {
3     private String title;
4     private String author;
5     private double price;
6     public Book(String title, String author, double price) {
7         this.title = title;
8         this.author = author;
9         this.price = price;
10    }
11    public String getTitle() { return title; }
12    public String getAuthor() { return author; }
13    public double getPrice() { return price; }
14    public String toString() {
15        return "Title: " + title + ", Author: " + author + ", Price: "
16    }
17    public static void main(String[] args) {
18        HashMap<String, Book> library = new HashMap<>();
19        library.put("978-1-2345-6789-7", new Book("Java Basics", "ABC",
20        library.put("978-0-9876-5432-1", new Book("Data Structures", "I
21        library.put("978-2-3456-7890-2", new Book("Algorithms", "GHI",
22        String isbnToSearch = "978-0-9876-5432-1";
23        if (library.containsKey(isbnToSearch)) {
24            System.out.println("Book found for ISBN " + isbnToSearch +
25            System.out.println(library.get(isbnToSearch));
26        } else {
```

Input

stdin input

Output

```
Book found for ISBN 978-0-9876-
5432-1:
Title: Data Structures, Author:
DEF, Price: 599.0
All books in the library:
```

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5/5 answered

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.

- (a) List three advantages of Generics in Collections.
- (b) Represent how employee data is stored using List with Generics.
- (c) Write a Java program using Iterator to:
- Store employee objects
 - Display employees with salary > 50,000

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.List;
4 class Employee {
5     private int id;
6     private String name;
7     private double salary;
8     public Employee(int id, String name, double salary) {
9         this.id = id;
10        this.name = name;
11        this.salary = salary;
12    }
13    public double getSalary() {
14        return salary;
15    }
16    public String toString() {
17        return "Employee{id=" + id + ", name='" + name + "', salary=" + salary + "}";
18    }
19    public static void main(String[] args) {
20        List<Employee> employees = new ArrayList<>();
21        employees.add(new Employee(1, "Alice", 45000));
22        employees.add(new Employee(2, "Bob", 60000));
23        employees.add(new Employee(3, "Charlie", 52000));
24        employees.add(new Employee(4, "Diana", 30000));
25        System.out.println("Employees with salary > 50000:");
26        Iterator<Employee> iterator = employees.iterator();
27        while (iterator.hasNext()) {
```

Input

stdin input

Output

```
Employees with salary > 50000:
Employee{id=2, name='Bob',
salary=60000.0}
Employee{id=3, name='Charlie',
salary=52000.0}
```


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Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- (a) Identify three suitable Collection types for this system.
(b) Design a structure using Set and Map for voters and vote counting.
(c) Write a Java program using HashSet and HashMap to:

- Record votes
- Prevent duplicate voting
- Display results

Your Code

```
1 import java.util.*;  
2 public class OnlineVotingSystem {  
3     public static void main(String[] args) {  
4         Set<String> votedUsers = new HashSet<>();  
5         Map<String, String> voterToCandidate = new HashMap<>();  
6         Map<String, Integer> candidateVotes = new HashMap<>();  
7         String[] candidates = {"Alice", "Bob", "Charlie"};  
8         for (String c : candidates) {  
9             candidateVotes.put(c, 0);  
10        }  
11        vote(votedUsers, voterToCandidate, candidateVotes, "V1", "Alice");  
12        vote(votedUsers, voterToCandidate, candidateVotes, "V2", "Bob");  
13        vote(votedUsers, voterToCandidate, candidateVotes, "V3", "Alice");  
14        vote(votedUsers, voterToCandidate, candidateVotes, "V2", "Charlie");  
15        vote(votedUsers, voterToCandidate, candidateVotes, "V4", "Charlie");  
16        System.out.println("\n Voting Results ");  
17        for (String candidate : candidateVotes.keySet()) {  
18            System.out.println(candidate + " : " + candidateVotes.get(candidate));  
19        }  
20        System.out.println("\n Voter Details ");  
21        for (String voter : voterToCandidate.keySet()) {  
22            System.out.println(voter + " voted for " + voterToCandidate.get(voter));  
23        }  
24    }  
25    static void vote(Set<String> votedUsers,  
26                    Map<String, String> voterToCandidate,  
27                    Map<String, Integer> candidateVotes,
```

Input

stdin input

Output

```
Vote recorded: V1 : Alice  
Vote recorded: V2 : Bob  
Vote recorded: V3 : Alice  
Duplicate vote rejected for  
voter: V2  
Vote recorded: V4 : Charlie
```